Prompt 2 - no MCP tool description - run 1 (seed 0)

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

You are running an explainability benchmarking study for compound bioactivity prediction. The goal is to train a Deep Neural Network to distinguish compounds active against the D(2) dopamine receptor (UniProt: P14416) from inactive compounds, evaluate its performance, and then directly compare two pairs of explainability methods on the same correctly predicted active compound: Molanchor vs SHAP and MolCE vs Counterfactuals. Summarize where the methods agree on the structural determinants of the predictions and where they disagree.

Assistant

(Calling tool `find_datasets`)

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(Calling tool `load_dataset`)

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(Calling tool `compute_features`)

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(Calling tool `split_dataset`)

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(Calling tool `get_top_k_bit_environments_from_shap`)

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(Calling tool `explain_with_molce`)

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(Calling tool `generate_counterfactuals`)

Assistant

🧪 **Explainability Benchmarking Study: D(2) Dopamine Receptor**

This study evaluates a Deep Neural Network (DNN) trained to distinguish active compounds (Class 1) from inactive ones (Class 0) for the D(2) dopamine receptor (UniProt: P14416). We then performed a direct comparison of explainability methods on a correctly predicted active compound.

1. Model Performance

The DNN was trained on the `chembl_activity_data_P14416_binary` dataset (8,348 samples, 70/30 split).

- **Test Accuracy:** 98.6%
- **Test AUC:** 0.996
- **Test Balanced Accuracy:** 98.6%

2. Selected Compound Analysis

We selected a compound from the test set that was **correctly predicted as active** with high confidence.

Property	Value
SMILES	`CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O`
True Label	1 (Active)
Predicted Label	1 (Active)
Confidence	1.0

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3. Explainability Comparison

We compared two pairs of methods: **Molanchor vs. SHAP** and **MolCE vs. Counterfactuals**.

Pair 1: Molanchor vs. SHAP

Both methods identify **local structural determinants** (substructures) that drive the prediction.

- **Agreement:** Both methods highlight the **piperazine-like core** and the **N-methyl** group as critical for activity.
  - **Molanchor** identified a specific anchor rule: `[*]N1CCC2(CC1)C(=O)N(C)CN2[*]` with 100% precision.
  - **SHAP** (Top Bits) identified environments such as `CCCN(C)C`, `CN(C)C`, and `cc(CN(C)C)c`, which correspond to the piperazine ring and its N-methyl substituents.
- **Disagreement:**
  - **Molanchor** provides a single, high-confidence logical rule (anchor) representing a substructure.
  - **SHAP** provides a granular, bit-level contribution map, revealing that specific atom environments (like the sulfur in the benzothiazole ring `cc(c)s`) also contribute significantly, which Molanchor's single anchor might not fully capture if it focuses only on the core.

Pair 2: MolCE vs. Counterfactuals

Both methods identify **what changes** would cause the prediction to flip (from active to inactive).

- **Agreement:** Both methods agree that the **piperazine core** and the **benzothiazole moiety** are essential for activity. Modifying these regions leads to a loss of activity.
  - **MolCE** identified contrastive R-groups like `*CC(C)NC(C)=O` that, if present, would likely reduce activity.
  - **Counterfactuals** generated alternative structures like `COCCCON1...` (modifying the core) or `CSC(=S)NN=CN1...` (modifying the heterocycle) that are predicted to be inactive.
- **Disagreement:**
  - **MolCE** focuses on **contrastive features** (R-groups) that explain *why* it is active by contrasting with inactive examples. It highlights specific chemical groups that "kill" activity.
  - **Counterfactuals** provides **complete synthetic alternatives** (full molecular structures) that are inactive. While MolCE says "remove this group," Counterfactuals says "replace this molecule with this other molecule."

4. Summary of Structural Determinants

Method	Key Structural Determinants Identified
Molanchor	**Piperazine-like core** (`N1CCC2...`) with N-methyl groups.
SHAP	**Piperazine core**, **N-methyl groups**, and **benzothiazole moiety** (specifically the sulfur environment).

Method	Key Structural Determinants Identified
MolCE	**Piperazine core** and **benzothiazole** are essential; replacing them with aliphatic or polar groups (e.g., `*CC(C)NC(C)=O`) reduces activity.
Counterfactuals	**Piperazine core** and **benzothiazole** are essential; substituting them with polar/charged groups (e.g., `COCCCON-`) or different heterocycles flips the prediction to inactive.

Conclusion

All four methods converge on the importance of the **piperazine-like core** and the **benzothiazole-like aromatic heterocycle** for D(2) receptor activity.

- **Molanchor** and **SHAP** agree on the *presence* of these features being necessary.
- **MolCE** and **Counterfactuals** agree on the *necessity* of these features by showing that their removal or modification leads to inactivity.

The main difference lies in the **granularity**: SHAP provides bit-level detail, Molanchor provides a logical rule, MolCE provides contrastive features, and Counterfactuals provides full structural alternatives.